

Week 11 – Faruq Pradhan

MCQ Item Analysis

1. The difficulty range is slightly below goal (should be > 0.92) given that the discrimination index is low and so there may be benefit in changing the distractors (either C or D since no one selected either option) to increase the difficulty slightly assuming this is a competency item. Q-index above goal 0.80
2. Under the presumption that this question is a discrimination item, the difficulty index is slightly high (should be between 0.27-0.73) with a discrimination index at goal. Consider revising the stem to decrease the difficulty of the question slightly. Q index above goal 0.80
3. Presuming this is also a discrimination item, the difficulty index at goal is in range; however, the discrimination index should be > 0.20 . The low Q index would also support changes to the multiple-choice question, specifically in the question stem (especially given the Q index < 0.65).

Psychometrics Short Answer

5. *The Kuder-Richardson 20 metric is a measure of internal reliability for an examination. (Google it for more information about how it is calculated and what desirable levels are.) What could a decreasing KR-20 for the same examination administered over time signify? (For example, a chemistry instructor gives the same 100-item exam to her group of CHEM 101 students every semester, and she notices over time that the KR-20 has steadily decreased.)*

A progressively decreasing score would imply decreasing reliability or internal consistent. This metric incorporates item score variance and a score of 0.70 or greater is considered reliable. In this context, reliability is typically defined as test-retest score similarity (i.e., whether there is a component of random error contributing) (Loudon & Macias-Muñoz, 2018). A decreasing score of time could suggest that the quality of the education delivered on specific topics may have changed over the years, or potentially a decrease in foundational knowledge (such as high school foundation for a 101-intro college course) amongst the student cohort. Alternatively, this could also be suggestive of a compromised examination due to the same exam being used every semester and individuals who took the test passing on exam info to incoming students as a form of academic dishonesty.

Loudon, C., & Macias-Muñoz, A. (2018). Item statistics derived from three-option versions of multiple-choice questions are usually as robust as four- or five-option versions: implications for exam design. *AJP Advances in Physiology Education*, 42(4), 565–575. <https://doi.org/10.1152/advan.00186.2016>

6. *Opportunities for student cheating are typically well-managed by large-scale assessors and instructors/proctors in academic settings, but students are creative creatures and find ways to undermine systems in place. Further, students naturally discuss test content with*

their classmates and those who come behind them in a program. These behaviors compromise the security of individual test items over time. How might an instructor use psychometric indices to identify items that should be phased out of an examination because they've been compromised either by direct acts of academic dishonesty or as the indirect result of student behavior?

Ranger et al., (2020) describe multiple different indicators to aid in delineating “irregular response processes”. The U1 statistic is based on the premise that a correctly answered difficult question implies that all easier questions have been correctly answered. Missed easier responses but correct difficult responses (Guttman errors) can be indicative of irregular responses processes. The U3 statistic is similar but includes a weighting of Guttman errors. A CS statistic includes covariance of responses within a specific pattern and compares this against Guttman scores. H^T marker incorporates a given response pattern to questions compared to the remaining response pattern in questions yet to be answered also using covariance. Other statistical measures to detect for irregular behaviors include indicators based on response times (such as the KL statistic or Z² statistic). Finally, Ranger et al., (2020) also provide statistical options to detect for irregular behavior based on the number of response revisions over the course of the test.

Ranger, J., Schmidt, N., & Wolgast, A. (2020). The Detection of Cheating on E-Exams in Higher Education—The Performance of Several Old and Some New Indicators. *Frontiers in Psychology, 11*. <https://doi.org/10.3389/fpsyg.2020.568825>

7. A student who has always been at the top of his class comes to you as a course director in the first year of a health sciences program. He is concerned because he is struggling with examinations in this program and wants some suggestions for how he can better prepare for the next examination. You only have the psychometric indices for the latest exam and are not a subject matter expert. The psychometrics report includes: difficulty index, discrimination index/point biserial correlation, class average, standard deviation, lowest and highest score. How would you use this information to advise this student?

It would be important to reference the difficulty index of the questions being asked to reassure the student that the test is intended to be difficult and discriminate between students that understand the content from those that don't. I would use the discrimination index to also reassure the student (assuming a high index) that the exams are designed to be challenging. His individual performance can also be used to compare against the mean and standard deviation to demonstrate his standing compared to his peers. Comparing one's score against the range without utilizing more information about the distribution would not be a truly accurate reflection of student performance.